# Averaging perfectly calibrated forecasters: a sharp $8/27$, a tight $1/2$, and the resolution in between

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## Abstract

It is folklore — made precise by Ranjan and Gneiting (2010) — that a nontrivial average of *distinct* perfectly calibrated probability forecasters is itself *not* calibrated, and is underconfident. That result is qualitative. We quantify it for binary outcomes under the $L^1$ calibration error (ECE). Three facts. (i) **Upper bound:** the average (any convex weights, any number of forecasters) has ECE $\leq 1/2$, with a two-line proof. (ii) **A sharp, interpretable witness:** two perfectly calibrated *Bayesian* agents who see disjoint information and predict $Y = A \vee B$ have an equal-weight average with ECE $= 2q(1-q)^2$, maximized at $q = 1/3$ to give exactly $\mathbf{8/27 \approx 0.296}$ — and its three reported values $\{1/3, 2/3, 1\}$ are spread out, so this miscalibration survives *any* binning. (iii) **The $1/2$ ceiling is tight, but only at infinite resolution:** we exhibit an explicit family of calibrated pairs whose average has population ECE $\to 1/2$ (the maximum a forecaster can have), so averaging can be *as miscalibrated as a forecaster can be* — yet the witnessing reports cluster into a vanishing band around the base rate, so at any fixed bin width their measured ECE collapses to $\approx 0$. The resolution-robust worst case lies strictly between $8/27$ and $1/2$; with $m$ averaged agents the disjoint-information construction climbs toward $1/2$ with *spread* reports (ECE $\approx 0.47$ at $m = 12$), so even at coarse resolution the average can be nearly maximally miscalibrated. A simulation verifies every number.

## 1. Setup and notation

Fix a probability space carrying a binary outcome $Y \in \{0,1\}$ and forecasters, each a random variable $P \in [0,1]$ jointly distributed with $Y$. A forecaster $P$ is **perfectly calibrated** if

$$\mathbb{E}[Y \mid P] = P \quad \text{a.s.} \tag{1}$$

i.e. among instances on which it announces value $v$, the event happens a $v$-fraction of the time. Its **(population) $L^1$ calibration error**, the infinite-sample, infinite-resolution ECE, is

$$\mathrm{CE}(P) \;=\; \mathbb{E}\big|\,\mathbb{E}[Y \mid P] - P\,\big|, \tag{2}$$

so $\mathrm{CE}(P) = 0$ iff $P$ is calibrated. In practice ECE is computed by partitioning $[0,1]$ into $b$ bins and comparing each bin's mean forecast to its event rate; (2) is the $b \to \infty$ (group-by-exact-value) limit. We will need both.

Given calibrated forecasters $P_1, \dots, P_m$ and convex weights $w_k \geq 0$, $\sum_k w_k = 1$, the **linear pool** (average) is $\bar{P} = \sum_k w_k P_k$. The question of this note: *how large can* $\mathrm{CE}(\bar{P})$ *be?*

A fact we use repeatedly:

**Fact 0.** For any $\sigma$-field $\mathcal{G}$, the forecaster $P = \mathbb{E}[Y \mid \mathcal{G}]$ is perfectly calibrated.

*Proof.* $P$ is $\mathcal{G}$-measurable, so $\sigma(P) \subseteq \mathcal{G}$ and, by the tower rule, $\mathbb{E}[Y \mid P] = \mathbb{E}[\mathbb{E}[Y \mid \mathcal{G}] \mid P] = \mathbb{E}[P \mid P] = P$. ■

So *every* "rational Bayesian" forecast — the conditional expectation given whatever the agent knows — is automatically calibrated. The constructions below are exactly such forecasts.

## 2. An upper bound

**Proposition 1 (the average cannot exceed** $1/2$**).** Let $P_1, \dots, P_m$ be perfectly calibrated and $\bar{P} = \sum_k w_k P_k$ a convex combination. Then

$$\mathrm{CE}(\bar{P}) \;\le\; \sum_k w_k\, \mathbb{E}[2P_k(1-P_k)] \;\le\; \tfrac{1}{2}. \tag{3}$$

*Proof.* By Jensen for conditional expectations, $\mathrm{CE}(\bar{P}) = \mathbb{E}\big|\mathbb{E}[Y - \bar{P} \mid \bar{P}]\big| \le \mathbb{E}|Y - \bar{P}|$. Linearity and the triangle inequality give $\mathbb{E}|Y - \bar{P}| \le \sum_k w_k\, \mathbb{E}|Y - P_k|$. Finally, conditioning on $P_k$ and using calibration (1) — given $P_k = v$, $Y$ is Bernoulli$(v)$ — yields $\mathbb{E}[\,|Y - P_k| \mid P_k = v] = v(1-v) + (1-v)v = 2v(1-v)$, hence $\mathbb{E}|Y - P_k| = \mathbb{E}[2P_k(1-P_k)] \le 2 \cdot \frac{1}{4} = \frac{1}{2}$. ■

Two remarks. The middle expression in (3) is twice the average **sharpness deficit** $\mathbb{E}[P_k(1-P_k)]$: vague forecasters (values near $1/2$) leave the most room for the average to go wrong. And $1/2$ is the largest ECE *any* binary forecaster can have whose reports lie in $[0,1]$ with the matching base rate; so Proposition 1 says the average is never worse than a worst-possible forecaster — the interesting question is how close it can get.

## 3. A sharp, interpretable witness: $8/27$

The mechanism behind miscalibration of the pool is **information diversity** (Satopää et al. 2014; Baron et al. 2014): two agents who each see *part* of the picture are each calibrated on their own slice, but neither value, nor their average, equals the conditional rate given *both* slices. The cleanest instance:

**Theorem 2 (disjoint-information OR).** Let $A, B \overset{\text{iid}}{\sim} \text{Bernoulli}(q)$ and $Y = A \vee B = \mathbf{1}\{A = 1 \text{ or } B = 1\}$. Let agent 1 report $P_1 = \mathbb{E}[Y \mid A] \in \{q, 1\}$ and agent 2 report $P_2 = \mathbb{E}[Y \mid B] \in \{q, 1\}$. Both are perfectly calibrated (Fact 0), yet their equal-weight average $\bar{P} = \frac{1}{2}(P_1 + P_2)$ has

$$\mathrm{CE}(\bar{P}) \;=\; 2\,q(1-q)^2, \qquad \max_q \;=\; \tfrac{8}{27} \approx 0.2963 \;\text{ at } q = \tfrac{1}{3}. \tag{4}$$

*Proof.* Since $A = 1 \Rightarrow Y = 1$ and $A = 0 \Rightarrow Y = B$, we get $P_1 = 1$ if $A = 1$ and $P_1 = \mathbb{E}[B] = q$ if $A = 0$; likewise for $P_2$. The average therefore takes three values, by the number of "on" bits:

| event | prob. | $\bar{P}$ | $\mathbb{E}[Y \mid \bar{P}]$ | \|gap\| |
|---|---|---|---|---|
| $A = B = 0$ | $(1-q)^2$ | $q$ | 0 | $q$ |
| exactly one on | $2q(1-q)$ | $\frac{1+q}{2}$ | 1 | $\frac{1-q}{2}$ |
| $A = B = 1$ | $q^2$ | 1 | 1 | 0 |

The middle row is the crux: if exactly one bit is on then $Y = 1$ with certainty, but the average splits the difference between the confident "1" and the cautious "$q$" and announces $\frac{1+q}{2} < 1$. Summing mass $\times$ gap,

$$\mathrm{CE}(\bar{P}) = (1-q)^2 q + 2q(1-q) \cdot \tfrac{1-q}{2} + q^2 \cdot 0 = 2q(1-q)^2.$$

Differentiating, $\frac{d}{dq}\, 2q(1-q)^2 = 2(1-q)(1-3q) = 0$ at $q = \frac{1}{3}$, giving $2 \cdot \frac{1}{3} \cdot \frac{4}{9} = \frac{8}{27}$. ■

Two things make this witness strong. First, it is **resolution-robust**: the reports are $\{1/3, 2/3, 1\}$, spread far apart, so the same ECE $= 8/27$ is recorded by *any* binning at $b \geq 3$ bins (the simulation confirms ECE $= 0.2963$ at 10 bins). Second, it is **exactly the underconfidence of Ranjan and Gneiting (2010)** made numerical: at $\bar{P} = \frac{1+q}{2}$ the truth is 1 (forecast too low) and at $\bar{P} = q$ the truth is 0 (forecast too high) — both errors point *toward* the base rate $\mathbb{E}[Y] = 1 - (1-q)^2$. The pool is systematically timid because each agent already hedged over what it could not see, and averaging hedges twice.

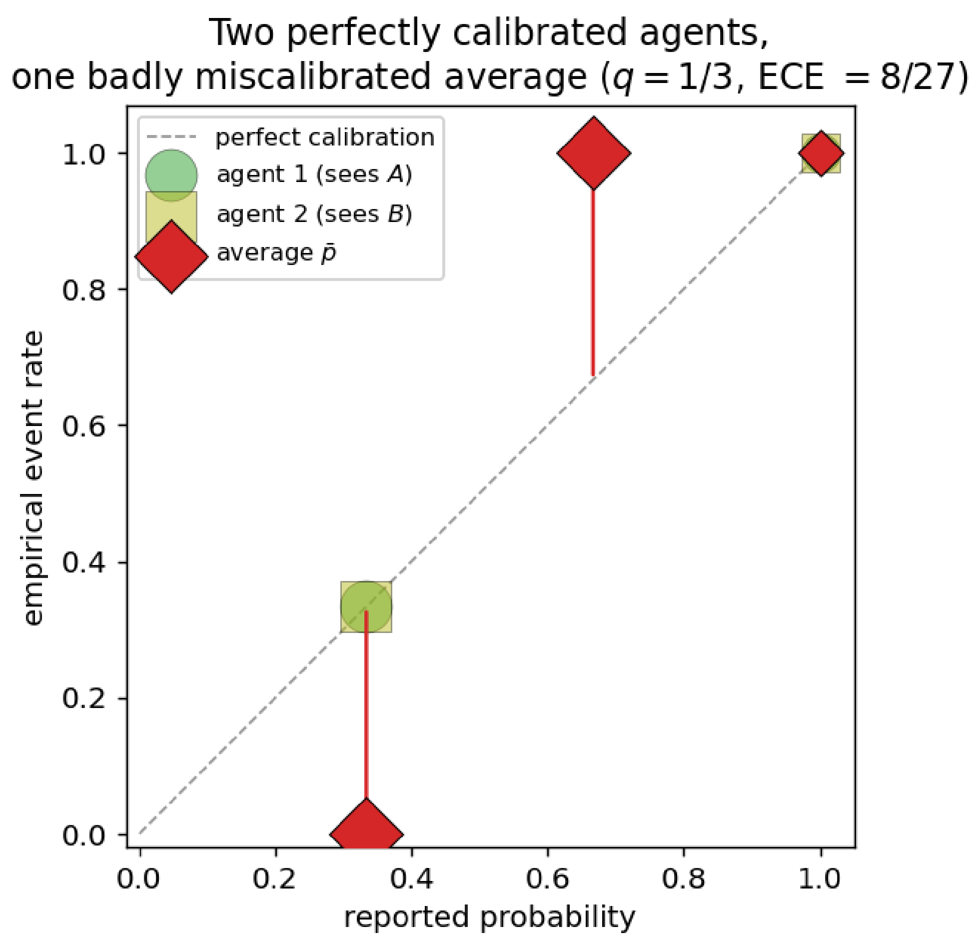


Figure 1: Two perfectly calibrated agents lie on the diagonal; their equal-weight average (red) sits far off it — at reported 1/3 the event never happens, at reported 2/3 it always does. Marker area is proportional to probability mass. Finite-sample ($n = 3 \times 10^5$): average ECE $= 0.2965$ vs population $8/27 = 0.2963$; each agent's ECE $\leq 0.001$.

## 4. The $1/2$ ceiling is tight — but it is a statement about resolution

Is $8/27$ the worst case? No — not even close, if we measure (2) literally.

> **Proposition 3 (tightness of the bound).** $\sup\ \mathrm{CE}(\bar{P}) = \frac{1}{2}$, the supremum over all pairs of perfectly calibrated forecasters with equal weights. It is approached but not attained.

*Construction.* Take a $2\times 2$ contingency table over $(i,j) \in \{0,1\}^2$ with cell rates $r_{ij} = \mathbb{P}(Y = 1 \mid \text{cell})$ set to the **checkerboard** $r_{00} = r_{11} = 1$, $r_{01} = r_{10} = 0$, and cell masses

$$\pi = \begin{pmatrix} \frac{1}{4} + p & \frac{1}{4} - p \\ \frac{1}{4} - 2p & \frac{1}{4} + 2p \end{pmatrix}, \qquad 0 < p < \tfrac{1}{8}.$$

Let agent 1 report the row-mean rate and agent 2 the column-mean rate; by Fact 0 both are perfectly calibrated. Their values are $a_0 = \frac{1}{2} + 2p,\ a_1 = \frac{1}{2} + 4p$ (rows) and $b_0 = \frac{1/4+p}{1/2-p},\ b_1 = \frac{1/4+2p}{1/2+p}$ (columns), all $\to \frac{1}{2}$ as $p \to 0$. The average $\bar{P}_{ij} = \frac{1}{2}(a_i + b_j)$ thus takes four *distinct* values, all approaching $\frac{1}{2}$, while each cell's outcome is deterministic ($r_{ij} \in \{0,1\}$). Each value is therefore its own singleton "bin" with gap $|r_{ij} - \bar{P}_{ij}| \to \frac{1}{2}$ and mass $\to \frac{1}{4}$, so

$$\mathrm{CE}(\bar{P}) \ \longrightarrow\ 4 \cdot \tfrac{1}{4} \cdot \tfrac{1}{2} = \tfrac{1}{2} \qquad (p \to 0).$$

Combined with Proposition 1, $\sup \mathrm{CE}(\bar{P}) = \frac{1}{2}$. The simulation traces it: $p = 0.1, 0.05, 0.02, 0.01, 0.005, 0.002, 0.001$ give $\mathrm{CE} = 0.306, 0.452, 0.492, 0.498, 0.4995, 0.4999, 0.500$. $\square$

So **averaging two calibrated forecasters can be as miscalibrated as a forecaster can be.** But read the construction again: as $p \to 0$ every report falls into a band $[\,\frac{1}{2} - O(p),\ \frac{1}{2} + O(p)\,]$ around the base rate. The forecasters become almost the constant "$\frac{1}{2}$" predictor — calibrated precisely because the base rate is $\frac{1}{2}$ — and only an *infinitely* fine reliability diagram can see that the four nearly-identical reports actually point at deterministic 0s and 1s. At any fixed bin width the reports eventually share a bin and the measured ECE vanishes. The simulation makes the dependence explicit (Figure 2): at $p = 0.01$ the population ECE is 0.498 while the 20-bin ECE is 0.000 and the 100-bin ECE is 0.005. By contrast the $8/27$ construction's reports are a fixed distance apart, so its ECE is identical at 20, 100, or infinitely many bins.

The honest worst case is therefore *resolution-indexed*. At infinite resolution it is $1/2$ (Proposition 3); the resolution-robust worst case — what a real reliability diagram with a fixed bin width can witness — is strictly smaller and exceeds $8/27$ (a $2 \times 2$ pair with well-separated reports already reaches ECE $\approx 0.45$, stable under $\geq 50$-bin binning; see `sim.py`). The two regimes are not in tension: $8/27$ is the cleanest robust *example*, not the robust *maximum*.

### More forecasters: a robust climb toward $1/2$

The clustering caveat is special to two agents. With $m$ disjoint-information agents — bits $A_1, \ldots, A_m \overset{\text{iid}}{\sim} \mathrm{Bernoulli}(q)$, $Y = \bigvee_i A_i$, agent $i$ reporting $\mathbb{E}[Y \mid A_i] \in \{q,\ 1 - (1-q)^{m-1}\}$ — the average's reports stay *spread out* (one value per number of "on" bits), so its ECE is resolution-robust, and it climbs toward the ceiling: optimizing $q$ per $m$ gives $\mathrm{CE} = 0.296, 0.372, 0.406, \ldots, 0.470$ for $m = 2, 3, \ldots, 12$, identical at 20 bins or at infinite resolution (Figure 3). So once you pool more

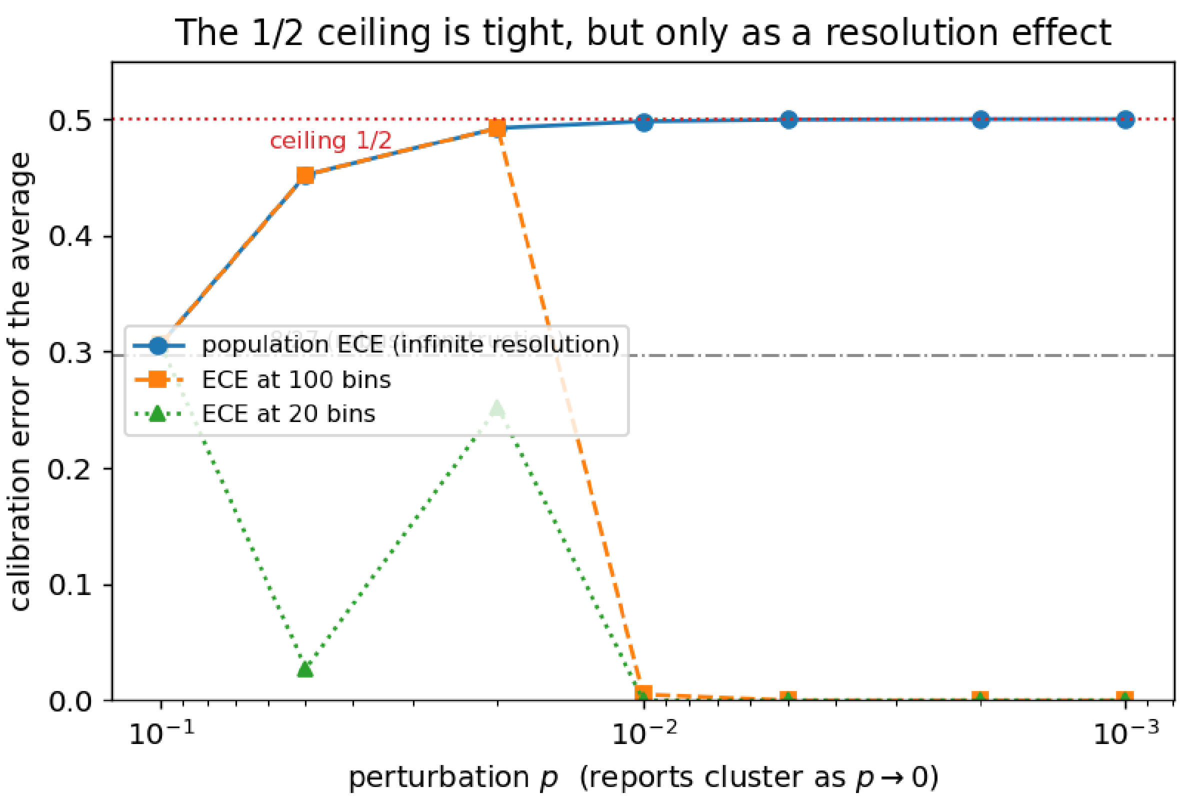


Figure 2: The $1/2$ ceiling is tight only at infinite resolution. As reports cluster ($p \to 0$, axis reversed) the population ECE climbs to $1/2$, but the fixed-resolution ECE collapses once all reports fall inside one bin. The $8/27$ construction (gray line) is robust at every resolution.

than a couple of diverse-but-calibrated forecasters, the average can be *robustly* close to maximally miscalibrated.

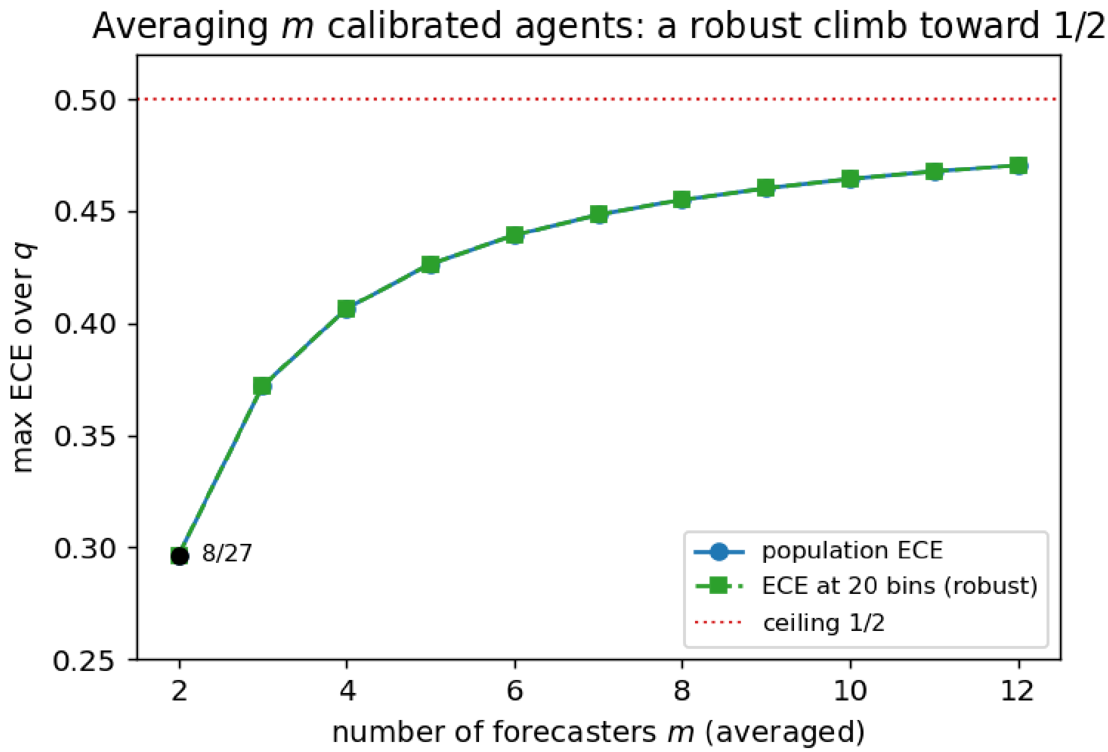


Figure 3: With $m$ averaged agents the disjoint-information construction keeps its reports spread, so population and 20-bin ECE coincide (resolution-robust) and climb toward the 1/2 ceiling. The $m = 2$ point is the 8/27 witness.

## 5. Discussion

**What is new.** That the linear pool of distinct calibrated forecasts is uncalibrated and underconfident is Ranjan and Gneiting (2010), in the real-valued / PIT setting; the "extremize the average" correction (Satopää et al. 2014; Baron et al. 2014) is the operational response. Those statements are qualitative. We give the binary-outcome magnitude: a tight ceiling $\mathrm{CE}(\bar{P}) \leq 1/2$ (Prop. 1), a clean closed-form witness 8/27 with an interpretable Bayesian story (Thm. 2), and the fact that the ceiling is tight (Prop. 3) but only at infinite resolution — making explicit that the "worst-case miscalibration of an ensemble" is not a single number but a function of the reliability diagram's bin width.

**Practical reading.** (i) Averaging probabilities is not a calibration-preserving operation, and the damage is not negligible: even two diverse calibrated models can land $8/27 \approx 0.30$ off, and several can approach 0.5. Recalibrate *after* pooling (Platt/isotonic/beta), or extremize. (ii) The resolution caveat doubles as a warning about ECE itself: the metric's verdict on a pooled forecaster can swing from "perfectly calibrated" to "maximally miscalibrated" purely by changing the bin width, which is exactly the binning sensitivity documented for ECE estimators (Guo et al. 2017; Vaicenavičius et al. 2019; Kumar et al. 2019). A pooled forecaster whose reports cluster near the base rate is the adversarial case where this matters most.

**Limitations.**

- *Binary outcomes, $L^1$ error.* Proposition 1's clean 1/2 uses $Y \in \{0, 1\}$ and the $L^1$ (ECE) error. The squared/reliability ($L^2$) version and multiclass or real-valued $Y$ (where Ranjan–Gneiting live) would have different constants; we did not compute them.
- $8/27$ *is a witness, not the robust maximum.* We did not determine the exact resolution-robust worst case at a fixed bin width $b$ (the value that increases from below toward 1/2 as $b \to \infty$). Our numerical search over $2 \times 2$ pairs suggests it exceeds 0.45 even at moderate $b$, but we state no theorem; that constant — and its dependence on $b$ — is left open.
- *$m \to \infty$ limit.* The $m$-agent ECE is still rising at $m = 12$ (0.470); we did not compute its limit (a Poisson-regime calculation as $mq \to \lambda$) and so do not claim it reaches 1/2.
- *Perfect calibration assumed.* We compare against *perfectly* calibrated inputs to isolate the effect of averaging; with empirically-calibrated inputs the pool inherits their error too.

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